Worksheet 6.3 — Cartesian Vectors in 2-D & 3-D

Grade 12 Calculus and Vectors (MCV4U) · Unit 6: Geometry and Algebra of Vectors

The Big Ideas

Component form (a, b) or (a, b, c); magnitude $\sqrt{a^2 + b^2 (+c^2)}$; operate componentwise; $\vec{AB} = B - A$; unit vector $\vec{v}/|\vec{v}|$.

- $|(a, b)| = \sqrt{a^2 + b^2}$, $|(a, b, c)| = \sqrt{a^2 + b^2 + c^2}$.
- Add/subtract/scale componentwise; $\vec{AB} = B - A$.
- Unit vector $\hat{v} = \vec{v}/|\vec{v}|$.

The Big Picture

Component form

A **Cartesian vector** is written by components, $\vec{v} = (x, y)$ or (x, y, z), making calculations algebraic rather than geometric.

Magnitude

The magnitude is $\sqrt{x^2 + y^2}$ in 2-D and $\sqrt{x^2 + y^2 + z^2}$ in 3-D the distance formula in vector form.

Operations by component

Add, subtract, and scale vectors component by component; a unit vector is the vector divided by its magnitude.

Worked Examples

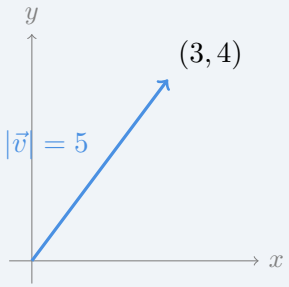
Example 1 — Magnitude (2-D)

Find $|(3, 4)|$.

Solution. **Step 1 (apply the 2-D magnitude formula):** $|(a, b)| = \sqrt{a^2 + b^2}$.

Step 2 (substitute): $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$.

Conclusion: $|(3, 4)| = 5$:



Example 2 — Magnitude (3-D)

Find $|(1,2,2)|$.

Solution. **Step 1 (apply the 3-D magnitude formula):** $|(a,b,c)| = \sqrt{a^2+b^2+c^2}$.

Step 2 (substitute): $\sqrt{1^2+2^2+2^2} = \sqrt{1+4+4} = \sqrt{9}$.

Conclusion: $|(1,2,2)| = 3$.

Example 3 — Componentwise

Compute $(2,3)+(1,-1)$ and $3(2,-1)$.

Solution. **Step 1 (add component by component):** $(2+1,\ 3+(-1)) = (3,2)$.

Step 2 (scale each component by 3): $(3\cdot 2,\ 3\cdot(-1)) = (6,-3)$.

Conclusion: $(2,3)+(1,-1) = (3,2)$ and $3(2,-1) = (6,-3)$.

Example 4 — Between points

Find $\vec{AB}$ and $|\vec{AB}|$ for $A(1,2), B(4,6)$.

Solution. **Step 1 (subtract A from B, tail from tip):** $\vec{AB} = B - A = (4-1,\ 6-2) = (3,4)$.

Step 2 (find the magnitude): $|\vec{AB}| = \sqrt{3^2+4^2} = \sqrt{25}$.

Conclusion: $\vec{AB} = (3,4)$, with length 5.

Example 5 — Unit vector

Unit vector of $(3,4)$.

Solution. **Step 1 (find the magnitude):** $|(3,4)| = \sqrt{9+16} = 5$.

Step 2 (divide each component by the magnitude): $\left(\frac{3}{5}, \frac{4}{5}\right)$.

Conclusion: the unit vector is $(\frac{3}{5}, \frac{4}{5})$.

Example 6 — Magnitude (2-D)

Find $|(6,8)|$.

Solution. **Step 1 (apply the magnitude formula):** $\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100}$.

Conclusion: 10.

Example 7 — Magnitude (3-D)

Find $|(2,3,6)|$.

Solution. **Step 1 (apply the 3-D magnitude formula):** $\sqrt{2^2+3^2+6^2} = \sqrt{4+9+36} = \sqrt{49}$.

Conclusion: 7.

Example 8 — Componentwise

Compute $(5,1)-(2,4)$.

Solution. **Step 1 (subtract component by component):** $(5-2,\ 1-4)$.

Conclusion: $(3,-3)$.

Example 9 — Between points

Find $\vec{AB}$ for $A(2,1), B(5,5)$.

Solution. **Step 1 (subtract A from B):** $(5-2,\ 5-1)$.

Conclusion: $\vec{AB} = (3,4)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $|(6, 8)|$.

Question 2

Find $|(2, 3, 6)|$.

Question 3

Compute $(5, 1) - (2, 4)$.

Question 4

Find $\vec{AB}$ for $A(2, 1), B(5, 5)$.

Question 5

Unit vector of $(6, 8)$?

Question 6

Find $|(5, 12)|$.

Question 7

Find $|(1, 4, 8)|$.

Question 8

Compute $(3, 2) + (4, -5)$.

Question 9

Compute $2(3, -4)$.

Question 10

Find $|\vec{AB}|$ for $A(0,0), B(8,6)$.

Question 11

Unit vector of $(0,5)$?

Question 12

Find $|(7,24)|$.

Question 13 — Challenge

Given $A(1,2,3)$ and $B(4,6,15)$, find $\vec{AB}$ and $|\vec{AB}|$.

Question Recap & Answer Key

Question Recap

1. Find $|(6,8)|$.
2. Find $|(2,3,6)|$.
3. Compute $(5,1)-(2,4)$.
4. Find $\vec{AB}$ for $A(2,1), B(5,5)$.
5. Unit vector of $(6,8)$?
6. Find $|(5,12)|$.
7. Find $|(1,4,8)|$.
8. Compute $(3,2)+(4,-5)$.
9. Compute $2(3,-4)$.
10. Find $|\vec{AB}|$ for $A(0,0), B(8,6)$.
11. Unit vector of $(0,5)$?
12. Find $|(7,24)|$.
13. Given $A(1,2,3)$ and $B(4,6,15)$, find $\vec{AB}$ and $|\vec{AB}|$.

✓ Answer Key

1. 10
2. 7
3. $(3,-3)$
4. $(3,4)$
5. $(\frac{3}{5},\frac{4}{5})$
6. 13
7. 9
8. $(7,-3)$
9. $(6,-8)$
10. 10
11. $(0,1)$
12. 25
13. $(3,4,12)$, $|\vec{AB}|=\sqrt{9+16+144}=13$